

Human Review: Asymptotic Dimension Zero ACL Counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 19 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed construction gives a correct negative answer to Problem 2.15 of Braga–Lancien, *Asymptotic coarse Lipschitz equivalence*, as the problem is stated for arbitrary metric spaces. The source question asks whether two asymptotically coarse Lipschitz equivalent metric spaces of asymptotic dimension zero must be coarse, or coarse Lipschitz, equivalent. The packet constructs spaces X and Y with

$$\text{asdim}(X) = \text{asdim}(Y) = 0,$$

which are asymptotically coarse Lipschitz equivalent but are not even coarsely equivalent. Since coarse Lipschitz equivalence implies coarse equivalence, this is the stronger negative conclusion.

The idea is clear. The space $X = \{x_n : n \in \mathbb{N}\}$ is a sparse sequence with

$$d_X(x_n, x_m) = 2^n + 2^m \quad (n \neq m).$$

The space Y replaces each point x_n by a fiber $\{n\} \times A$, with internal distance n , while keeping different fibers at distance $2^n + 2^m$. Collapsing each fiber gives a coarse Lipschitz map $g : Y \rightarrow X$, and choosing one representative in each fiber gives a coarse Lipschitz map $f : X \rightarrow Y$. One has $gf = \text{Id}_X$, while fg moves a point in the n -th fiber by at most n . Since this is sublinear compared with the distance scale 2^n , the maps witness asymptotic coarse Lipschitz equivalence.

The asymptotic dimension zero verification is also sound. At a fixed scale r , the finitely many small fibers may be grouped into one bounded set, while for all sufficiently large n , the points inside the n -th fiber are already farther than r apart, so singletons handle the large fibers.

The proof that the spaces are not coarsely equivalent is correct. If a coarse equivalence existed, there would be coarse maps $F : X \rightarrow Y$ and $G : Y \rightarrow X$ with FG uniformly close to Id_Y . Thus there is $C < \infty$ such that

$$d_Y(FG(y), y) \leq C \quad (y \in Y).$$

Choosing a fiber whose internal distance is larger than C forces FG to be the identity on that fiber. In the packet's original version, where the fiber is uncountable and X is countable, this gives an immediate cardinality contradiction.

Suggested Countable Variant

The uncountability of the set A is not actually needed. A more natural version of the example is obtained by taking

$$A = \mathbb{N},$$

so that both X and $Y = \mathbb{N} \times \mathbb{N}$ are countable. The asymptotic coarse Lipschitz equivalence and the asymptotic-dimension-zero argument are unchanged.

Only the final non-coarse-equivalence argument needs a small adjustment. If $F : X \rightarrow Y$ and $G : Y \rightarrow X$ were coarse maps with $FG \sim \text{Id}_Y$, let C be the closeness constant and choose $n > C$. Then FG must be the identity on the infinite fiber

$$Y_n = \{n\} \times \mathbb{N}.$$

Hence

$$F(G(Y_n)) = Y_n.$$

But Y_n has finite diameter n . Since G is coarse, $G(Y_n)$ has bounded diameter in X . Every bounded subset of the sparse space X is finite, so $G(Y_n)$ is finite, and therefore $F(G(Y_n))$ is finite. This contradicts the infinitude of Y_n .

Thus the example can be made countable. It still does not have bounded geometry or locally finite fibers, so it should still be viewed as answering the literal formulation of the source problem rather than any stronger bounded- geometry version.

Review Outcome

Archive this packet as a correct literal-wording counterexample to Problem 2.15. For presentation, I recommend replacing the uncountable fiber set by the countable choice $A = \mathbb{N}$, together with the adjusted finite-image argument above. This makes the example more natural while preserving the same obstruction.